

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

28th November 2019

Subject CB2 – Business Economics

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q.1)** Rational Decision making requires that:
- A. One's choice be arrived at logically and without error
 - B. One's choice be consistent with one's goals
 - C. One's choices never vary
 - D. One makes choices that do not involve trade-offs
- [1.5]**
- Q. 2)** Which of the following would not result in a rightward shift of the Production Possibility curve?
- A. An increase in investment in Capital Stock.
 - B. A reduction in the labour unemployment rate.
 - C. The discovery of new oil deposits in India.
 - D. An increase in the number of people taking management training courses.
- [1.5]**
- Q. 3)** If the marginal product of labour is below the average product of labour, it must be true that:
- A. The marginal product of labour is negative.
 - B. The marginal product of labour is zero.
 - C. The average product of labour is falling.
 - D. The average product of labour is negative.
- [1.5]**
- Q. 4)** Which of the following issue relates to micro-economics?
- A. Impact of crude price hike on inflation
 - B. Impact of change in bank rate on bank saving and investment
 - C. Impact of shortage of wheat production on wheat prices
 - D. Impact of Information technology on economic growth
- [1.5]**
- Q. 5)** What is the difference between the term 'customer' and the term 'consumer'?
- A. There is no difference.
 - B. The term consumer refers only to people who buy food and drink products.
 - C. Customers buy products but it is consumers who use them.
 - D. Customers make organisational rather than personal purchases.
- [1.5]**
- Q. 6)** Price discrimination in which seller charges less to customers, who buy in large volumes is classified as
- A. Second-degree price discrimination
 - B. First-degree price discrimination
 - C. Third-degree discrimination
 - D. Fourth-degree discrimination
- [1.5]**

- Q. 7)** The classical economists believed that
- A.** The Government should intervene in order to reach full employment.
 - B.** Wages and salaries suffer from certain rigidities and would thus not move in a direction that would ensure full employment.
 - C.** A competitive economy free of Government intervention would result in full employment.
 - D.** An economy in equilibrium could also be an economy with high unemployment. [1.5]
- Q. 8)** Which of the following is the correct definition of the law of diminishing returns?
- A.** If extra units of one variable input are added to a fixed amount of all other inputs, then sooner or later the marginal returns will get smaller.
 - B.** If extra units of one variable input are added to a fixed amount of all other inputs, then the marginal returns will always get smaller.
 - C.** If extra units of all variable inputs are added to a fixed amount of all fixed inputs, then the marginal returns will always get smaller.
 - D.** If extra units of all variable inputs are added to a fixed amount of all fixed inputs, then sooner or later the marginal returns will get smaller. [1.5]
- Q. 9)** Which of the following is not an argument against Government intervention in a market?
- A.** If the Government reduces output below the free market level, consumer surplus will fall.
 - B.** If the Government reduces output below the free market level, producer surplus will fall.
 - C.** If the Government raises output above the free market level, consumers will get no benefit from the extra units of output.
 - D.** If the Government raises output above the free market level, some units of output will cost more to produce than the value placed on them by consumers. [1.5]
- Q. 10)** Suppose the Government introduces a specific tax of Rs 5 a unit on a product which has sloping supply and demand curves. Which of the following statements about the resulting curve $S+tax$ is true?
- A.** The gap between $S+tax$ and S will be higher at high quantities than at low quantities.
 - B.** The gap between $S+tax$ and S will be higher at low quantities than at high quantities.
 - C.** The gap between $S+tax$ and S will be Rs 5 at each quantity.
 - D.** The gap between $S+tax$ and S will be whatever is needed to show that the equilibrium price for consumers increases by Rs 5. [1.5]
- Q. 11)** A consumer finds that for product A, the price is Rs 5 and the consumer's marginal utility is 100 utils, while for product B, the price is Rs 10 and the consumer's marginal utility is 160 utils. Which of the following statements is true?
- A.** The consumer is maximizing utility from A and B.
 - B.** The consumer would gain more utility from A and B by consuming more A and less B.
 - C.** The consumer would gain more utility from A and B by consuming less A and more B.
 - D.** The consumer could only gain more utility from A and B by consuming more of both products. [1.5]

- Q. 12)** Suppose you give up a job and use some savings to buy a small bus which you drive on a remote rural route. Which of the following costs of your firm is an explicit cost?
- A. The depreciation on the bus.
 - B. The wage of the job you gave up.
 - C. The fuel you buy for the bus.
 - D. The interest you earned on your savings before you used them to buy the bus.
- [1.5]**
- Q. 13)** Which of the following is NOT TRUE of a risk averse person?
- A. A risk averse person chooses a course of action designed to maximise expected utility.
 - B. A risk averse person will tend to reject a fair gamble.
 - C. A risk averse person will be unwilling to pay more for insurance than the long run average value of claims which will be made.
 - D. A risk averse person will be willing to pay more for insurance than the long run average value of claims which will be made.
- [1.5]**
- Q. 14)** An increase in the central bank's target rate of inflation is represented by a
- A. Movement up the Dynamic Aggregate Demand curve.
 - B. Movement down the Dynamic Aggregate Demand curve.
 - C. Rightward shift in the Dynamic Aggregate Demand curve.
 - D. Leftward shift in the Dynamic Aggregate Demand curve.
- [1.5]**
- Q. 15)** According to the Phillips curve, unemployment will return to the natural rate when:
- A. Nominal wages are equal to expected wages
 - B. Real wages are back at long-run equilibrium level
 - C. Nominal wages are growing faster than inflation
 - D. Inflation is higher than growth of nominal wages
- [1.5]**
- Q. 16)** Overshooting models of the exchange rate are an attempt to explain:
- A. Why purchasing power parity plays no role in determining the value of a currency
 - B. Difference in inflation rates between the countries
 - C. Why foreign exchange market is never in equilibrium
 - D. Why exchange rates are so volatile
- [1.5]**
- Q. 17)** If the exchange rate is below the equilibrium level then in a floating exchange rate system:
- A. There is excess demand and the exchange rate should fall.
 - B. There is excess supply and the exchange rate should fall.
 - C. There is excess demand and the exchange rate should rise.
 - D. There is excess supply and the exchange rate should rise.
- [1.5]**

- Q. 18)** In the European Union:
- A. All member countries have a single exchange rate
 - B. All member countries have set their own tariff
 - C. There is a common tariff against non-members
 - D. All taxes are set the same
- [1.5]**
- Q. 19)** The IS curve illustrates that when real national income increases, the
- A. Interest rate must fall to restore equilibrium in the money market.
 - B. Interest rate must rise to restore equilibrium in the money market.
 - C. Interest rate must rise to restore equilibrium in the goods market.
 - D. Interest rate must fall to restore equilibrium in the goods market.
- [1.5]**
- Q. 20)** Suppose a central bank is concerned about inflation and wishes to raise interest rates. Which of the following instruments would it be inappropriate to use?
- A. Open market operations
 - B. Interest rate control
 - C. Reserve ratio controls
 - D. Quantitative easing
- [1.5]**
- Q. 21)** What is the main reason that led classical economists to believe that the Government need not respond to a demand shock which led to unemployment?
- A. Government always act very late
 - B. Demand shocks are always very small
 - C. Real wages are highly flexible
 - D. Economies never displayed demand-deficient unemployment
- [1.5]**
- Q. 22)** If private investment increased by \$150 billion while GDP remained the same, which of the following could have occurred, all else being the same?
- A. Consumption spending decreased by \$150 billion
 - B. Exports increased by \$150 billion
 - C. Imports decreased by \$150 billion
 - D. Net exports increased by \$150 billion
- [1.5]**
- Q. 23)** When there is unemployment caused by a lack of consumer spending, we classify this type of unemployment as?
- A. Seasonal
 - B. Structural
 - C. Cyclical
 - D. Regional
- [1.5]**

- Q. 24)** An example of an automatic stabilizers within the economy is
- A.** Changes in Government spending
 - B.** Tariff on imports
 - C.** A subsidiary approved by the Government
 - D.** Changes in spending for unemployment compensation [1.5]
- Q. 25)** The trough of a business cycle occurs when _____ hits its lowest point.
- A.** Aggregate economic activity
 - B.** Inflation
 - C.** The unemployment rate
 - D.** National income [1.5]
- Q. 26)** An increase in (i) the price level and (ii) the rate of inflation:
- A.** Cause the demand for money (i) to rise; (ii) to fall
 - B.** Cause the demand for money (i) to fall; (ii) to rise
 - C.** Both cause demand of money to fall
 - D.** Both cause demand of money to rise [1.5]
- Q. 27)** i) Given below are four statements.
- A.** Air Vivanta reduces its average plane fares by 30% in order to attract more passengers.
 - B.** The Government grants export subsidy to producers of oranges in Nagpur to increase the sale of oranges abroad.
 - C.** Rice farmers decide to withhold rice as the market price is low.
 - D.** OPEC decides to increase the international oil price.
- For each of the above statements, choose whether it reflects increase or decrease in
- demand,
 - quantity demanded,
 - supply,
 - quantity supplied. (2)
- ii)** Consider the following market structures. Show the demand curves and their elasticities. Include diagrams in the explanations:
- a)** Perfect Competition (2)
 - b)** Monopolistic Competition (2)
 - c)** Monopoly (2)
- [8]**

Q. 28) Discuss:

- i) Four exceptions to the law of demand. (4)
 - ii) Are all inferior goods necessities? Are all necessities inferior goods? Explain. (3)
- [7]**

- Q. 29)** i) Explain why the firm in an oligopoly market has no incentive to raise or cut price when it faces a kinked demand curve. (3)
- ii) Draw a diagram to show the profit maximising price and output of a firm with a kinked demand curve in an oligopoly market. (2)
- [5]**

Q. 30) Consider that Dan produces pen in a perfectly competitive industry. The following are his costs:

Quantity	TC (Rs.)
0	50000
1000	55000
2000	58000
3000	59000
4000	64000
5000	70000
6000	83000
7000	99000
8000	122000
9000	149000
10000	200000

- i) What is Dan's Breakeven Price? (2)
 - ii) What is Dan's shut down price? (2)
 - iii) Suppose the price of a pen is Rs 7. What is the profit maximising quantity of pens that Dan should produce? Will he produce or shut down in the short run? Will he stay in the industry or exit in the long run? (3)
 - iv) Draw a diagram showing a profit maximising monopolist making a loss at its loss minimising output but still finding it worthwhile to produce in the short run. Include the following curves, marginal revenue (MR), average revenue (AR), average total cost (AC), average variable cost (AVC) and marginal cost (MC). Show the price (P), output (Q), and the level of average total cost (C) and average variable cost (D), at the loss minimising output. (4)
- [11]**

Q. 31) i) Define the following:

- GDP Deflator
- Consumer Price Index

(2)

- ii) State any two major differences between GDP deflator and Consumer Price index. (2)
- iii) Gross Domestic Product (GDP) in India grew 5% in the first quarter of FY20, marking the slowest growth since the fourth quarter of FY13. In your opinion, what is the list of key drivers responsible for the slowest GDP growth rate. (3)

[7]

- Q. 32)** i) Draw diagrammatic representation of Securitization arrangement. (2)
- ii) Explain role of securitization in financial crisis of 2008. (4)
- iii) Describe the fiscal policies adopted by many Governments in response to the financial crisis and its aftermath. (4)

[10]

- Q. 33)** i) Explain with the help of Taylor Rule, how Central bank of the country achieves its economic objectives of target inflation and target level of Real GDP. (4)
- ii) Define dynamic aggregate demand curve (DAD) and explain how action of Central Bank affects the slope of dynamic aggregate demand curve. (2)

[6]

- Q. 34)** i) Assume prices and interest rates are fixed. An economy is described by the following equations:

$$\text{Consumption} = 100 + 0.6 Y$$

$$\text{Investment} = 100$$

$$\text{Government spending} = 100$$

$$\text{Export} = 40$$

$$\text{Import} = 40 + 0.1 Y$$

Where, Y = National Income

Find the equilibrium output, consumption, imports, and the trade deficit. (3)

- ii) Define J curve effect. (1)
- iii) List the drivers for the potential globalization of an industry. (1)
- iv) Use the IS-MP model to illustrate and explain the effect on interest rates and national income of an increase in the demand for exports. (2)

[7]
